



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Autumn Semester Examination 2022-23

Date of Examination: 21/11/2022

Session: AN

Duration: 3 hrs. Full Marks: 50

Subject No.: PH31207

Subject: Classical Mechanics

Department/Center/School: Physics

Specific charts, graph paper, log book etc., required

Special Instructions (if any): Answer all questions

1. Runge-Lenz vector

- (a) Define the Poisson's bracket between two phase space quantities $\{A, B\}_{qp}$.
- (b) Evaluate the Poisson's brackets between the following quantities
 - i. $\{r_i, L_j\}$
 - ii. $\{p_i, L_j\}$
 - iii. $\{L_i, L_j\}$
 - iv. $\{r_i, L^2\}$

where the angular momentum is defined as usual $\vec{L} \equiv \vec{r} \times \vec{p}$ and $L^2 = \sum_i L_i^2$

- (c) The Runge-Lenz vector is defined as

$$\vec{R} = \vec{p} \times \vec{L} - \frac{\vec{r}}{r}$$

show that

$$\{L_i, R_j\}_{qp} = \epsilon_{ijk} R_k$$

where ϵ_{ijk} are the usual antisymmetric tensors.

- (d) For a free particle with $H = p^2/2m$ show that the Runge-Lenz vector is conserved

[2+(2+2+2+2)+5+5]

2. Harmonic Oscillator and Canonical Coordinate Transformation

- (a) Consider a Harmonic Oscillator with a Hamiltonian given by

$$H(p, q) = \frac{p^2}{2m} + \frac{1}{2}kq^2$$

Given a generating function

$$F(q, Q, t) = \frac{1}{2}m\omega q^2 \cot Q$$

that drive a Canonical Coordinate Transformation between $[q, p, H] \rightarrow [Q, P, H']$. answer the following questions:

- i. Write down q and p as a function of Q and P .
- ii. Derive the form of H'
- iii. Identify the cyclic coordinate in H'
- iv. Solve the Hamilton's equation for Q and P
- (b) Show that Canonical Transformations form a group.
- (c) Does the Poincare recurrence theorem violate the 2nd law of thermodynamics. Explain

[(4+2+2+4)+4+4]

3. Consider a 1 dimensional system evolving in a phase space spanned by $(p_x - x)$, under the influence of a potential $V(x) = mgx$.

- (a) Construct the Hamiltonian of the system
- (b) Write down the Hamilton-Jacobi equation for the system
- (c) Derive the W-function by solving the H-J equation using separation of variables
- (d) Solve for the equation of motion using the H-J algorithm.

[2+2+3+3]